Prime Factorization, Divisor Sums, and Arithmetic Functions

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The elementary statement that every positive integer factors uniquely into primes has surprisingly different consequences: it gives algorithms, product formulas for divisor counts, inversion formulas, and eventually Euler products. A useful thread is that prime exponents turn multiplication into addition, divisor sums into algebraic convolutions, and finally arithmetic functions into generating series.[1]

§1.1 A divisor below a square root

If an integer $n > 1$ is composite, write $n = ab$ with $1 < a \le b$. Then $a^2 \le n$, hence $a \le \sqrt{n}$. Factoring a supplies a prime divisor $p \le \sqrt{n}$. Thus it suffices to test primes no larger than $\sqrt{n}$ to decide whether n is prime.

For $n = 221$, it is enough to test the primes $2, 3, 5, 7, 11, 13$, because $\sqrt{221} < 15$. The factor 13 succeeds:

$$221 = 13 \cdot 17.$$

Trial division and the sieve of Eratosthenes exploit the same bound in different ways: the sieve marks composites in a whole interval, whereas trial division tests one integer.

§1.2 Euclid's theorem and the prime-divisor lemma

Theorem 1.2.1 (Euclid)
There are infinitely many prime numbers.

Proof. If $p_1, \dots, p_k$ were all the primes, the integer $N = p_1 \cdots p_k + 1 > 1$ would have a prime divisor q. But $N \equiv 1 \pmod{p_i}$ for every i, so q is absent from the supposedly complete list. □

To obtain *uniqueness* of prime factorization, we need another fact.

Lemma 1.2.2 (Euclid's lemma)
If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

Proof. If $p \nmid a$, then $\gcd(a, p) = 1$. Bezout's identity supplies integers u, v with $ua + vp = 1$. Multiplying by b gives $uab + vpb = b$, whose left side is divisible by p because $p \mid ab$. Hence $p \mid b$. □

Repeated division proves the existence of a prime factorization. To prove uniqueness, compare two factorizations of the same integer. Euclid's lemma forces the first prime in one factorization to divide a prime in the other and hence to equal it. Cancel it, and continue by induction.

[1] Compare MIT 18.781, *Theory of Numbers*, lecture notes.

§1.3 Prime exponents as arithmetic coordinates

For $n > 0$, let $v_p(n)$ denote the exponent of a prime p in n. Unique factorization is the assertion that

$$n = \prod_p p^{v_p(n)}, \qquad v_p(n) \geq 0,$$

with finitely many nonzero exponents. It turns multiplication and divisibility into coordinatewise statements:

$$v_p(mn) = v_p(m) + v_p(n), \qquad a \mid b \iff v_p(a) \leq v_p(b) \quad \forall p.$$

Consequently the greatest common divisor and least common multiple satisfy

$$v_p(\gcd(a,b)) = \min(v_p(a), v_p(b)), \quad v_p(\operatorname{lcm}(a,b)) = \max(v_p(a), v_p(b)).$$

For example, $72 = 2^3 3^2$, $90 = 2 \cdot 3^2 \cdot 5$, so

$$\gcd(72,90) = 2 \cdot 3^2 = 18, \qquad \operatorname{lcm}(72,90) = 2^3 3^2 5 = 360.$$

§1.4 Counting divisors and their sum

If $n = \prod_{i=1}^r p_i^{a_i}$, every divisor has the unique form $d = \prod_i p_i^{b_i}$ with $0 \leq b_i \leq a_i$. Choosing each b_i independently gives

$$\tau(n) = \sum_{d \mid n} 1 = \prod_{i=1}^r (a_i + 1).$$

Summing the values of all choices instead gives

$$\sigma(n) = \sum_{d \mid n} d = \prod_{i=1}^r (1 + p_i + \cdots + p_i^{a_i}) = \prod_i \frac{p_i^{a_i+1} - 1}{p_i - 1}.$$

For $n = 12 = 2^2 \cdot 3$, the divisors are $1, 2, 3, 4, 6, 12$, so $\tau(12) = 6$ and $\sigma(12) = 28$. The formulas agree:

$$(2+1)(1+1) = 6, \qquad (1+2+4)(1+3) = 28.$$

A positive integer n is a perfect square exactly when all its $v_p(n)$ are even; by the formula for τ, this is equivalent to $\tau(n)$ being odd. Likewise, n is called perfect when $\sigma(n) = 2n$, as for $n = 6$.

§1.5 Legendre's formula is a double count

The exponent of p in $n!$ is

$$\boxed{v_p(n!) = \sum_{j \geq 1} \left\lfloor \frac{n}{p^j} \right\rfloor.}$$

Each multiple of p contributes at least one factor; each multiple of p^2 contributes one additional factor, and so on. For example,

$$v_2(10!) = \lfloor 10/2 \rfloor + \lfloor 10/4 \rfloor + \lfloor 10/8 \rfloor = 5 + 2 + 1 = 8.$$

This also yields a practical test for binomial coefficients:

$$v_p\binom{n}{k} = v_p(n!) - v_p(k!) - v_p((n-k)!).$$

The integrality of $\binom{n}{k}$ is reflected in the nonnegativity of every one of these prime exponents.

§1.6 Dirichlet convolution encodes divisor sums

An arithmetic function assigns a complex number to each positive integer. For such functions define

$$(f * g)(n) = \sum_{d \mid n} f(d) g(n/d).$$

The function $\varepsilon(1) = 1$, $\varepsilon(n) = 0$ for $n > 1$ is the identity under convolution. If $\mathbf{1}(n) = 1$ and $\mathrm{id}(n) = n$, then

$$\tau = \mathbf{1} * \mathbf{1}, \qquad \sigma = \mathbf{1} * \mathrm{id}.$$

The first identity counts one ordered pair of divisors d, e with $de = n$ for each divisor d. The second weighs each pair by e; it computes precisely the divisor sum.

Associativity is worth checking rather than assuming. Both $(f * g) * h$ and $f * (g * h)$ sum $f(a)g(b)h(c)$ over the same ordered triples $abc = n$.

§1.7 Möbius inversion removes previous divisor sums

Define the Möbius function by

$$\mu(n) = \begin{cases} 1, & n = 1, \\ (-1)^r, & n \text{ is the product of } r \text{ distinct primes}, \\ 0, & p^2 \mid n \text{ for some prime } p. \end{cases}$$

For $n > 1$ with r distinct prime factors,

$$\sum_{d \mid n} \mu(d) = \sum_{j=0}^{r} \binom{r}{j} (-1)^j = (1-1)^r = 0.$$

For $n = 1$ the sum is 1. Thus

$$\mu * \mathbf{1} = \varepsilon.$$

If $F(n) = \sum_{d \mid n} f(d) = (\mathbf{1} * f)(n)$, convolution by μ gives the *inversion formula*

$$f(n) = \sum_{d \mid n} \mu(d) F(n/d).$$

Example 1.7.1

The integers $1 \le k \le n$ with $\gcd(k, n) = 1$ are counted by Euler's totient $\varphi(n)$. Partitioning $1, \dots, n$ by their gcd with n gives

$$\sum_{d \mid n} \varphi(d) = n.$$

Möbius inversion yields

$$\varphi(n) = \sum_{d \mid n} \mu(d) \frac{n}{d} = n \prod_{p \mid n} \left(1 - \frac{1}{p}\right).$$

For $n = 12$, this predicts $\varphi(12) = 12(1 - \tfrac{1}{2})(1 - \tfrac{1}{3}) = 4$, matching the units $1, 5, 7, 11$ modulo 12.

§1.8 Euler Products and Factorization

For $\mathrm{Re}(s) > 1$, the series

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$$

converges absolutely. Expanding geometric series for every prime p and using unique factorization gives

$$\boxed{\zeta(s) = \prod_{p} (1 - p^{-s})^{-1} \qquad (\mathrm{Re}(s) > 1).}$$

The region of convergence matters: no unrestricted interchange of infinitely many sums and products has been justified outside it.

The simple congruence exercise $2, 4, 8, 6 \pmod{10}$ belongs to another consequence of the same arithmetic structure: units modulo m form a finite multiplicative group. A periodicity claim about powers must therefore be checked against coprimality. For example, 2 is not a unit modulo 10, so its observed eventual cycle is not an instance of Euler's theorem on $(\mathbb{Z}/10\mathbb{Z})^{\times}$.

The progression is from a local divisibility test to unique prime coordinates, then to product formulas and inversion. Each new operation is motivated by a concrete earlier calculation; analytic continuation or deeper zeta-function claims can be studied separately.